PATRICK BASTARD

Backend Developer | AI Engineer (Junior)

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H-1 Visa (Valid until Feb 2027) | lilixe.github.io | linkedin.com/in/p-bastard/

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SUMMARY

Backend and AI-oriented software engineer with a Master's degree in Digital Imaging and Software Engineering. Experienced in building full-stack applications using FastAPI, SQLAlchemy, PostgreSQL/SQLite, and Streamlit, with a strong focus on data extraction, automation, and matching systems. Developed real-world projects involving job scraping, skill extraction, scoring algorithms, and database-backed tracking dashboards. Currently completing the Google Data Analytics Professional Certificate to strengthen analytics and business intelligence capabilities. Seeking entry-level or junior roles in backend engineering, AI, or data analytics.

PROFESSIONAL EXPERIENCE

Full Stack Engineer Intern, *Cooper Standard* — 02/2023 – 08/2023, France

- Built internal digital workflow tools to support industrial factory organization and production processes.
- Developed interactive web applications using JavaScript, HTML, and CSS.
- Collaborated with stakeholders to gather requirements and deliver usable tools for real business workflows.
- Improved internal efficiency by digitizing and simplifying manual processes.

Car Mechanic (Full-time/Contract), *Ronceray, Artis Interim* — 10/2023 – 08/2025, France

- Diagnosed and repaired complex mechanical systems with strict safety and quality requirements.
- Applied structured troubleshooting methods and delivered consistent technical results under time pressure.
- Installed vehicle options and equipment for resale preparation.
- Worked in a fast-paced environment requiring precision and reliability.

Remote Monitoring & Assistance Operator (Part-time), *Cegedev* — 07/2020 – 02/2023, France

- Monitored alarm systems and responded to emergency events in real time.
- Maintained accuracy, responsibility, and calm decision-making under pressure.

Web Developer Intern, *Solid'R* — 07/2021 – 08/2021, France

- Built and deployed a modern responsive website using Flutter.
- Improved UI usability and overall user experience.

Bartender / Waiter, *Volunteer* — Rennes, France

- Helping my friends running their bars/nightclubs from time to time

EDUCATION

Master's Degree – Digital Imaging & Software Engineering, — 09/2018 – 10/2023
ESIR (École Supérieure d'Ingénieurs de Rennes) — France

- Relevant Coursework: Deep Learning (PyTorch), AI for Image Applications, Backend Development (FastAPI/PostgreSQL), Algorithms, OpenGL Graphics, Unity Game Programming.

CORE SKILLS

Programming Languages
Python, C++, Java, C#, JavaScript

Backend Development
FastAPI, REST APIs, SQLAlchemy, CRUD, Web Scraping

Databases
SQLite, PostgreSQL

Data / Analytics
Skill extraction, scoring systems, data parsing, structured pipelines

AI / Machine Learning
Neural Networks fundamentals, backpropagation, MNIST training

Web / Frontend
Streamlit dashboards, HTML/CSS (basic)

Tools
Git/GitHub, VS Code, Docker

Other Domains
Digital Imaging, Computer Graphics (OpenGL), Unity, Teamwork, Communication

LANGUAGES

French
Native

English
Business, C1

Korean
Basic, learning

PROJECTS

Job Match Dashboard — Job Scraping & Skill Matching Platform, *Tech Stack: Python, FastAPI, SQLAlchemy, SQLite/PostgreSQL, Streamlit, Requests, BeautifulSoup*

- Built a full-stack platform to automatically scrape job postings from Wanted Korea and store results in a database.
- Extracted structured job data: title, company, URL, and description (tasks + requirements).
- Implemented a skill extraction engine using keyword pattern detection for relevant technical skills.
- Designed a scoring algorithm to compute match scores (%) comparing job requirements with user skills.
- Developed database-backed job tracking with status management (new/unfit/applied).
- Built a Streamlit dashboard with filtering tools, direct job link navigation, and one-click status updates.
- Added resume parsing functionality to extract skills from uploaded PDFs.
- Demonstrated strengths in backend development, REST APIs, database design, automation pipelines, scraping, data processing, and product-driven engineering.

NeuralNetwork — Neural Network Implementation in C++,
Tech Stack: C++, Machine Learning Fundamentals, MNIST

- Implemented a C++ feedforward neural network, emphasizing core ML mechanics.
- Developed training logic with forward and backward propagation and gradient-based weight updates.
- Built MNIST dataset loading and training pipeline to validate learning.
- Demonstrated understanding of machine learning workflows without reliance on libraries.
- Showcased strengths: algorithmic thinking, math-driven programming, performance-oriented implementation, and ML fundamentals.

SlaSHR — Visual-Inertial SLAM Implementation on MuSHR Robot, 09/2022 – 01/2023
Collaboration with DGA, Tech Stack: ROS1, C++, Python, Intel RealSense (D435i/T265), RViz, Evo

- Conducted a state-of-the-art study of open-source SLAM-VI algorithms and selected ORB-SLAM3, Kimera, and XIVO based on ROS compatibility, documentation quality, and real-time feasibility.
- Implemented and configured ORB-SLAM3 and Kimera-VIO within a ROS environment, adapting launch/config files to match MuSHR sensor topics and RealSense camera parameters.
- Built a custom indoor dataset by acquiring visual + IMU data from MuSHR RealSense cameras, covering multiple scenarios (corridors, staircases, lighting variations, loop closure trajectories).
- Solved sensor integration constraints by merging accelerometer and gyroscope streams into a unified IMU ROS topic required by ORB-SLAM3.
- Evaluated SLAM trajectory accuracy using Evo metrics (APE/RPE), comparing visual-only vs visual-inertial SLAM, and loop-closure vs non-loop trajectories.
- Demonstrated that IMU integration significantly improves localization stability and reduces drift, while environmental conditions (lighting, texture uniformity, staircases) strongly impact SLAM performance.
- Identified hardware/software limitations of the MuSHR Jetson Nano platform impacting real-time deployment and proposed improvements including migration from ROS1 to ROS2.
- Contributed technical documentation and installation guides to the project GitHub, supporting reproducibility and future development.
- https://github.com/mgodineau/SlaSHR#

CERTIFICATIONS

ETS TOIEC English Certification (C1)
Score 970